

Monopile Installation Record HB-A05

Harrow Bay Offshore Wind · Foundations

Elena Márquez · 05 May 2026 · HB-CON-IR-A05

Position	Date	Vessel	Contract
HB-A05	03 May 2026	MV Sea Atlas	WE-CON-2025-040 / -027

1. Position and pile data

Item	Design	As-installed
Coordinates (WGS84)	53.92040 N, 1.09820 E	within 11 cm of target
Water depth (m LAT)	30.6	30.8
Pile length (m)	74.6	74.6
Pile mass (t)	1,311	–
Target penetration (m)	33.9	34.1
Verticality	≤ 0.25°	0.19°

2. Driving record

Item	Value
Start of soft start	15:33
Driving duration	3 h 59 min
Total blow count	4,978
Maximum hammer energy	2,712 kJ
Self-weight penetration	6.2 m
Interruptions	1 × 25 min (hammer hose check)

3. Noise and marine mammal mitigation

Item	Value
Bubble curtain	Double big bubble curtain, both rings operational
Peak SEL at 750 m	158.1 dB (limit 160 dB)
Pre-piling search (MMO + PAM)	30 min, marine mammal detected – start delayed 20 min
Acoustic deterrent device	Activated 30 min before soft start

4. Transition piece

Item	Value
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TP installed	05 May 2026
Flange flatness after installation	0.9 mm (limit 2.0 mm)
Bolt tensioning	148 × M72 – completed
TP level	+0.01° (limit 0.25°)

5. Non-conformances

Observation: gripper pad wear close to limit, pads replaced after installation.

6. Sign-off

Role	Name	Date
Offshore Installation Supervisor	Atlas Marine Contractors	04 May 2026
Client Representative	Felix Brandt	04 May 2026
Reviewed	Elena Márquez	05 May 2026